

ActInf Textbook Group ~ Cohort 1 ~ Meeting 11

(Chapter 5 pt. 2)

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all right hello everyone it is week 11. and we're in chapter five part two of the active textbook group first cohort will be now turning to some of the more direct questions on chapter five after having more of a overview last week we'll also um be concluding our regular chapter discussions for the first part of this book and in the coming two weeks we'll be having some review and synthesis and connecting some dots asking overarching questions returning to questions that were based in a specific chapter and hopefully people like on their own or wherever else they're seeing is relevant or requesting new pages if they see something that they want providing some synthesis on this first half of the textbook which is the epistemic half and then um for people who want to continue in the textbook group go to the onboarding page and there's two columns there's yes for part two of cohort one chapter six through ten so check that box if you want to be included in starting in September going through chapters 6 through 10 which has like the recipe for active models and that's when we're going to be getting more Hands-On with the modeling also everyone is welcome to join for part one chapters one through five cohort two also starting in September so it could be um a fun experience just to get another coat of paint on the material play more active role ask another set of questions informed by what you've seen in the first cohort so feel free to um check those boxes and then also [Music] keep incubating as we head into chapter um six and Beyond in the second half of the textbook um like the project ideas several um of the project ideas like related to what Mike and Noah have added we've started to address and explore in the active block runs meetings on Wednesday especially related to like cyber physical systems simulations taxonomies for governance these kinds of ideas um Brock's added some great notes about like what what is the ramp that gets someone to the low road or the high road um and then uh I've recorded the first chapter and the first segments as an audiobook the first chapter doesn't have any equations so it's straightforward to read but the subsequent chapters do have equations which is um motivating having the natural language representations of the

equations so there's still a ton of um discourse questions that can be asked why things are one way why things are a different way if it can be placed into the equation itself that's great if it needs to be somewhere else hopefully the space exists and people can request if not but um for example when reading the textbook the equations can be read this way so those are just some of the projects that people can look forward to and the affordances to continue on which is again uh going to the onboarding and participating in first five chapters again cohort two or going to continue on with chapters six through ten starting in September for both of these so that we can like complete one pass of the book by the end of the year and also probably make a lot of progress on different project ideas and project ideas arising as we start to see and build templates for the models and also like Ali has some nice interactive notebooks so there will be a lot of great things um coming together if anyone wants to raise their hand of course just go for it otherwise we will be continuing with the questions asked about chapter five before we go to the questions is there any general comment that somebody wants to add about chapter five the reading of it they're rereading of it listening to the video from last week how is chapter five hitting them today okay um to kind of maybe bring that one step closer to a specific question I'm not going to read this whole quote but this question says in the preface they wrote in chapter 5 we will move from formal treatments to biological implications of active inference this AIDS in mapping the abstract computational principles of active inference to specific neural computations that can be executed by physiological substrates this is important in forming hypotheses under this framework and ensures that these are answerable to measure data in other words chapter 5 sets out the process Theory associated with active inference is this how people saw chapter 5 do they feel like it did something different did it succeed at making the abstract computational principles linked or mapped to specific neural computations executed by yes Eric you know first of all I have to apologize I only watched half of last year's last week's discussion so I may be cut talking about ground that's been covered before um it just seems to me that you know I've seen many many I um papers um trying to map theories onto brain circuitry and physiology and it's I I'm not well enough um you know I spent you know I'm only loosely associated with neuroscience and you know don't follow it very closely so it's hard for me to cite specific instances along the way I know you mentioned numanta which is um you know one of the more recent ones it's trying to say this is these are what the layers are doing um so this chapter to me didn't seem um any very qualitatively different from anybody else's very hand-wavy way to say here's what the computations are all about and here here's how they map to uh what brain circuitry could map

to what base circuitry might be doing it's all quite hand wavy I I find and I guess plausible but um it's certainly not definitive nothing close to that so maybe one other one other comment I would say is um again in this chapter they use this word prediction in a way that conforms to the theory but to say that you know what what's happening in um say um motor control where top-down signals from the cortex go down to the ganglia in the in the spinal cord in order to invoke motor patterns to call that a prediction is to me a kind of um well that's that's taking Extreme Measures to map the concepts of the theory onto what neurons are doing and um you know if the theory works out to be you know very useful in some way and that's what you have to do to to make that happen okay I guess I can use that language but it operatorly it just does not seem like the right language to use to say yeah we're trying to to set up some motor patterns set up some some context and situations for what we want the you know the local reflexes and and to be doing um but to cast that as a prediction just seems a little unnatural to me so those are my two reactions to this thank you anyone else with a raised hand this is a really good point about what the linkage or mapping is between uh without going into like you know is anything physical or Etc like physical brains in their Niche and any type of equations would a well-fitting linear model say that the brain is a linear model does a well-fitting base graph imply that the brain is a base graph does a well-fitting base graph plus a philosophical belief that we're just making a map we're not describing the territory equate to what Beyond a linear model or a verbal model Rohan yeah building on the previous point right uh I am still unsure what what uh active inference is considering a signal there's no effective well there's no noise model or there's it it doesn't specify that here's because if you if you say uh that I'm predicting something right so there must be something that is substantially different statistically different from some noise distribution that you can differentiate right so it's basically trying to estimate some hidden probabilities which is not really Sigma does it make sense because yours even the internal States unless they're actually uh instantiated prior like when we set up the model we say okay these are the only states that you can actually Traverse it does not make you still have to infer the States from your measurements of reality foreign yeah I just wanted to ask if the noise isn't kind of implied in um the pomdp formalism like uh I think in figure 5.4 where it shows the direct and invert Pathways with message passing it describes active in the basic ganglia as a umdp generative model so if it's partially observable doesn't that imply some kind of noise uh my counter to this would be uh that is missing information that needs to be filled out that's not really noise rate uh so noise would be something like if we uh like it so even if we let's let's not use this right it would be some so we

have like returns of a stock and looking at the distribution of the prior returns we know what the spread is and in case we want to measure whether your portfolio is substantially different from this distribution that's where we would have some sort of like sharp Ratio or a signal to noise ratio noise ratio being the complete set of distributions that are possible this set of ranges that are possible uh that is not really here because it that is missing information so we can only observe part the fact that we have only partial observations does not excuse the fact that we don't have uh even noise models of the sensors so how good are the sensors for example I do we have some idea of that yes but then I'll add a point about these Bayesian terminology yeah um yes um is there is there any like what's the Practical difference between partial observation where there's where the noise implies some kind of information that is missing versus like a distribution that's added to full observation that adds noise to like the sensory processing great so it so that would be some okay so if I use an actual physical sensor right uh it would be something like what are the ranges of values that it can go up to because we've designed the sensor so we would we basically know what the bias is what the variance of the sensor is under different conditions uh there are actual physical processes implied in that measurement so in the case of uh ultrasonic sensors probably like some current value or some voltage value that's being measured by some circuitry somewhere so there are ranges for that so it there's no missing information there so you basically know so it's already prior in this system right so you know what the ranges of values are going to be and from there you need to infer the some state in the world so is there an obstacle in the way or is there yeah okay let me just make a comment about some basic terminology then um Jessica and Brock so let's look at figure 4.3 this is the partially observable uh Markov decision process that Jacob talked about but we're going to be just talking about the perceptive part not the action intervention into how things are changing get just focusing on this Motif on the left like step-by-step guide uses so um you mentioned about the the bias related to measurements so if the measurables are the observable sense States then the bias and the covariance and the variance about how like the reading on the thermometer relates to the underlying variable the the hidden State the temperature of the room it's embodied in the a matrix because that is actually the learned or fixed variable that is playing a role of a noise model then you mentioned when talking about Returns on stocks testing whether there's like a substantial difference from some null expectation like is there a signal that's different from the noise um this is like a frequentist approach it's based upon determining whether there's some statistically significant level doesn't have to be it's not the only way to do it um but one can do like a p-value on whether a

given signal is statistically different from given a noise model different from noise and then like the sharp ratio and other types of things are like summary statistics they're like descriptive statistics noisy measurements are addressed in the context of the pomdp by having an a matrix that accommodates some of the features of denoising a signal and then Precision variables and those precisions can again be learned or fixed but like the functional aspects of what is being referred to as are accommodated within this kind of Bayesian approach which is why common filtering generalized Bayesian filtering why these techniques are used because we can have a Bayesian update scheme implemented with with variational approaches message passing sampling all these things that does denoise signal and it does embody a noise model and a signal model and that's all part of the generative model of the cognitive entity so it just isn't being addressed from a frequent to statistics or yeah I shouldn't have brought up the sharp ratio maybe that's not the best but the sharp ratio is uh is effectively some way to say that some portfolio choice is different from others or in communication systems you would see use something like EB over n_0 that's the signal noise ratio for any sort of communication system so yeah yeah uh but they're not necessarily only frequentest right so we there are patient ways to estimate these things in fact that's pretty much how we do it in practice because we we don't assume uh that we know the uh complete distribution we update priors in each of these cases because we have yeah uh coming back to your point that the a matrix does address this the a matrix would be something that would you said it would denoise and filter out some stray observations but okay so our so the the question Still Remains is when you're saying prediction is a prediction of the signal over here as in what is the future state of the sensor but I think there it seems seemed to indicate that the prediction relates to relates to the free energy minimization itself right so you choose policies that would minimize your free energy and keep yourself into some set point so which means that there should be some sort of uh an idea of what ranges are there for this value or uh how far can these go before you know you're not in that set point anymore so even if you if you do want to make that prediction it's a it's not just the signal value so signal values effectively through the Marco blanket they affect some internal State and we're that's what I I'm assuming that's what we're trying to predict the trajectory of the future internal States actually the expectation that's being um yeah minimized is over outcome observations not about internal States oh okay yeah yeah I'm sorry so uh yeah so even then we we don't have so the signals may have well-defined precision and bias but the entire system itself doesn't have any sort of well it doesn't know how wrong it could go it doesn't have any error estimate so that it's it there's no noise model as such so we can't apply

any so we would have to assume that there is some distribution of states and then use something like least squares or some other uh optimization routine to actually bring this in practice great totally possible um it would be interesting to sketch out a Bayes graph for what you believe a signal to noise or some type of noise model would embody and maybe we will see some analogies or some mappings um Jessica and then Brock hi yes I'm I mean I think I both you and Rohan like in the new Province but I was going to say that um would be like the filtering and like a lot of like the biases and things like that kind of like made sense of it is like I see the because I think it was like chapter two or something like that I said like right like so the why is this you know quote-unquote um you know uh like a fat thing like you know that we can observe from the hidden state or like the process of something that we don't know and so for everybody like we can see the same thing but how we process it internally is going to vary like it's gonna be very subjective so the why is gonna look different for Daniel then it's gonna look for me based on all this filtering and so the black Estates I imagine sort of like one half like those filters that encode the biases and all the things that allow us to process things differently and and so like then I generated model and it would like it's gonna be updated like differently all right also like encoded I think in the general model would be a lot of uh like you reset and things like that that embody I think probably like a lot of biases and like things of how we understand things which also influence how we interpret what we observe and so for so for me it's like a lot of these things are there like in terms of like statistics I don't know um much but that's how like I you know conceptually I um but like you know understood like a lot of these things so like basically like in the blanket stage of the Marco blanket it would have a ladder like this filtering you know and yes like um like if you're going to try to you know let's say Like You observe something that's different than what you predicted in your model then you would take action and to forage and like find information and to figure it out okay like what am I missing like why did I make this mistake or like this error I in my prediction and so that I can like have more information and update the model so that it aligns more closely so then you would basically take action to the noise and basically get closer to um like quote unquote the reality that's out there that yeah I don't know if it's official lighting but that's how like it kind of makes sense to me thanks Jessica Brock um yeah I just wanted to I guess speak to this like uh sensor noise model I don't like in practice in reality like when you're talking about sensors like in a physical system like like a digital signal processing kind of context like the sensors are tested under some conditions for things like you know Jitter and um spurious kind of uh noise conditions and stuff like that but they're tested under some

conditions meaning that there's never ever going to be a time when you have an actual model you're just going to have an approximation whether that's expressed in a frequentist or Bayesian way like there's going to be some uncertainty in the model and especially if you start connecting it up to a larger system um doing the same kind of like is the Jitter in that sensor the same when you know when it's completely isolated as it is when it's on a you know six by nine PCB board with you know 12 volts on one side and an oscillator next to that that's you know like are those now is it the same thing you know um you have to do these sort of electromagnetic compatibility you know resonance sort of testing on it like there's just that it's it's uh in practice it's Bayesian like you're doing the same sort of um bounding and you know belief um testing and you know Finding Your errors and correcting them um even in practice I don't see how you could possibly have like even like physically a system like that could exist where you had a noise model that you could write down that was actually the noise model um under all conditions or yeah it's um it doesn't seem like an approachable I don't know it's not even not tractable which is the wrong thing to do I don't thanks Brock um in the case of engineering and designing models it's up to the designer or the engineering team to understand like what is adequate or not and for natural systems outside of their bounds natural selection sweeps them off the table so some of the adequacy questions are either based upon specified or implicit human standards or the failure to resist dissipation and therefore the failure to realize repeated measurements of oneself or from the external and then that system is no longer let's just continue to move through some of the questions um yes go for it uh yeah I I completely agree with what Brock said right but my my point was when when you say that you're predicting something it means that uh you either know the quantity that you're trying to predict and or and it corresponds to so in this case like some action results in some reward I guess you could predict the reward but without knowing how wrong you're going to be in the future where exactly immigrant is so that any living system would have to have be aware of its own bounds right that's that's what I meant by noise models so it has to know that it has to stay within some bounds and as it starts coming closer to one of the upper lower bounds there should be some something like pulling it back down or some feedback mechanism but that is not very clear from the from what I've read so far on active inference system what how exactly it would estimate that is you can just save free energy minimization uh but what is a minimum in this case that would be different for different systems so it should have some idea of where the surfaces of whichever local surface it's lying on that that was my point so for sure it's different for different systems it's different as every single generative model is um and then

you mentioned like um a pullback attractor and so this has been treated extensively in the context of physiological measurements and then again wear those priors on the physiological measurements are the tolerable range are either provided by the human engineering team or through evolution by natural selection and then um yeah the quantity like another related topic would be expectation maximization models which is what is happening essentially in the dialectic between the observation and the hidden State update is an expectation maximization like process where it's like given the hyper priors or we can just simplify just given the priors on S what are the most likely or the distribution of observables and then given the observables what should be updated about the hidden State priors and when those are at conversions the model is stationary and then as things change either from the top via learning or context shift or through changed measurements the expectation maximization algorithm is just able to track those changes and that can in the context of a fixed prior have the function of a pullback attractor um okay and that's even before getting into action specifically but yes definitely considering like the role of comparing future plans of action and which plans of actions will have the least expected free energy based upon their um pragmatic and epistemic value like which the epistemic just being Clarity around how it's going to be achieved and the pragmatic value being that KL Divergence between the expectations preferences and the expected observations so if we expect slash prefer to be in homeostasis the pragmatic value is going to come or be loaded onto policies that keep us in homeostasis even if some other one is super informative how that model gets tuned is quite literally the details of how the model is trained right my point was that active inference doesn't help us find this we have to find this ourselves and then hopefully once we have instantiated these bounds I think active inference works very well to keep it within this right if you can't discover this you know right like I can't have like a uh what you'd say double or rasa type system like a blank slave system God in the world and discover your bounds that's not going to happen but this kind of method is what what I was uh making it out whereas taking some something equivalent without any so deep learning is another uh method right so you just feed it a lot of data it doesn't have to know anything about the data eventually it forms some opinion about the data it's seen and then it's able to do some classification or depending on how much data it's seen so it's able to build certain amount of bounds whether that's generalizable is a different question but uh here there's the learning part does not say much about how exactly it goes about discovering where its limits are that that's my sure thank you um Eric um you know I would I would love to to dive this a little bit more into um the thing you were just talking about how expectation maximization might is a is a

mathematical formulation for how to infer belief States from observations because you uh basically go through this iterative process of trying to find alignment between the states of the belief and the observables and then you can track that over time between that concept and the concept of message passing when I think of message passing I think of some sort of a distributed system where you have some local computation and it sends you know and you have again either synchronous or Asynchronous process for communicating between these kind of different centers of cut of belief state and then again through message passing you have often an iterative process of convergence so um those are not the same thing and I think you know message passing to my understanding would be an em algorithm under certain types of messages and States at each of the nodes that you're sending the messages around but not all message passing is going to be any em algorithm and similarly when I think of em I don't think about message passing I think about making an arrays of you know of um belief and matrices between you know mapping between belief and observations and then um trying to compute an expectation for you know for for what my predictions would be and then making this iterative conversion so I don't think of that in terms of message passing so I'd like to I just love to hear other people's ideas about how to tie these two um ideas together um Brock if your hand is still raced or okay nope so sorry yeah okay all good does anyone have thoughts on this there's a lot to say about the uh expectation maximization algorithms and like just to kind of give a little context this is the step-by-step um paper so this is the model stream one it's four parts um and this is like really a quite relevant figure which is not in the textbook but helps a lot like this is the essence of the partially observable Bayesian inference there's a prior D and then there's an ambiguity Matrix that is mapping in a generative capacity between hidden States s and observation measurements o this is The Tail of Two densities because o through a can give you the most likely s and S through a can give you the most likely o and it turns out that by alternating those procedures in a Bayesian update context that two-stroke engine is called expectation maximization again because given expectations on the summary statistics of a distribution the expected sensory observations can be generated and then given those observations a likelihood function can be maximized that then updates the hidden States so this Motif this kind of elbow motif is then extended into a caterpillar with this B which is how the hidden State changes their time so importantly like note what's not there which is like the observations being chained Through Time so it's not that the temperature reading at one moment influences the temperature reading at the next moment the temperature readings are continually linked um synchronously like in one time slice to their hidden States

so then when action gets into the picture we've talked about like several sources that's like the equation 2.5 to equation 2.6 phase change because action brings in several kinds of uncertainty first off you're reducing your uncertainty about observations which haven't happened yet you're also having the unknown consequences of your actions and in order to have action selection that's relevance one has to have a preference distribution over what kinds of observations they would like to be seeing so that action can guide it in that direction um then so that's the the expectation maximization two-stroke engine is basically graphically um proposed here D is just the initiating set of parameters and then in a One-Shot way like you can have a folder with a thousand images and do expectation maximization on it or you could have something that's dynamical and do that type of expectation maximization Through Time Eric asked how is it related to on graphs so I'm not familiar with all the details of this model but you basically said it which was that like certain message passing systems Implement em but of course not all message patching architectures Implement DM so message passing is more General than em also em wouldn't have to be implemented through message passing so there's kind of but there's like an area of intersection where the EM algorithm can be seen as a type of message passing under certain compute rules in a factor graph and EM at least from here may be used to break Cycles in a factor graph so it's always like how close to the kernel and like the platonic ideal of active inference are we talking and then how many heuristics and just ways of connecting things are possible so I I will now read the chat that's from previous context okay okay like the statistical parametric mapping textbook and documentation SPM has a lot on em variational inference and a lot of the parts that bring one to understand this a lot better SPM is like almost like it doesn't include action that much because it's a neuroimaging so the observations are neuroimaging Sensor Fusion different error modalities like the whole panopoly and then SPM started to incorporate participant actions in this partially observable metabasian way and potentially that is what led Tristan colleagues towards a grander synthesis of inference and action under a statistically principled framework let's just see if there's any other these are all you know important questions and like there's a lot to get up to them and then there's like a lot to go from the question but these are all important things to raise if anyone has a thought on this how should we think about redundancy in neural systems in the context of active so redundancy would be like if there's something that's playing a functional role but it's removal is not damaging the function of the system yeah there's 10 pillars knocking out one of them the building stays up so there's redundancy in the pillars what about the cases where the same neural circuits serves multiple

functions so how do we deal with the fact that there's like a many-to-many potentially or a complex mapping between system elements and system functions where sometimes removing Node 1 does nothing but removing node one and two there's a lot of things but then removing one two and three and it's going back to being fine Jessica and maybe like ask blue about it because this is what I feel but it this question made me think about like it's not only like that the brain has like this redundancy things but it like would like to say like if you have some kind of damage to your brain like another part of your brain that maybe you was not use for something would develop that capacity and so it would create that you know ability and so neuroplasticity like brand new even though it was not a redundancy to begin with so I guess I'm more Curious to understand this and so other people who know more like blue and stuff um about this and basically that's kind of what this question made me think about and kind of wanting to understand more that's good question yeah blue wrote depends on your age yeah like difference neuroplasticity mechanisms are differentially available throughout life for different organisms Just one thought on this would be like the same neural circuit serving multiple functions might be totally the case for example um and one like is the function of a heart to provide one pound of weight to the Torso it is a function of the heart is at one you know so identifying what function is being modeled is what's being done here so let's just go to like the example of figure five three this neuron modifying it like a loss of function experiments or having it injured might influence more things than just the lower motor neuron descending message so that'd be a case where like it serves multiple functions this is just one statistical model of this function of this circuit and that returns to to the earliest comments of Eric with like okay this is like tantalizingly seeming like it's actually going to be describing the neural Anatomy but then they say things like note the absence of dot dot dot dot note that there's a discrepancy between for example the reality of the anatomy and the base graph this highlights that the connections implied by message passing schemes may not manifest as single synapses there's kind of false positives and false negatives the Bayes graph isn't just the anatomy that's fine the linear regression between height and weight isn't the actual relationship between height and weight and the structural equation model of inequality is not the generator of inequality so it's totally fair and shouldn't be expected to be otherwise that the base graph the best fitting base graph the most didactic phase graph the simplest phase graph none of those recapitulate the anatomy and the base graph that recapitulates the anatomy would not necessarily even be the best fitting its map and territory and if somebody has a a special equation to break through that everybody would love to see it however sometimes it's easy to

gloss over that in principle challenge of mapping formalisms to biological systems because we see the cell and then the blanket is the membrane and we see the brain and it's like vision and action and it seems like it maps on to the physical or the anatomical structure of the world or the causal structure of the world and um it doesn't so I hope people can add more thoughts about like how we consider redundancy because it's a great question but we'll leave it there um how are reflexes modeled in active inference would they exist in an active inference module that is distinct from proprioception but they operate on different time scales um if anybody wants to add like some context on it a reflex arc relating to um where reflex is defined as um a function that is being relayed through the spinal cord of a mammal and not passing like you could have a nerve block in the cervical vertebra and it still is able to like Implement like that's one definition of reflex often it's also used slightly more broadly to mean like um stimulus action reproducible outcomes but not every reflex has a perfect reproducibility et cetera et cetera so it's kind of a Continuum um yes it would involve proprioceptive input that proprioceptive input would be combined or juxtaposed with the descending resulting in an error and then that directionality and magnitude of the error in this model drives the reflex there can also be multiple time scales but we haven't seen any nested modeling yet this is just the one layer model the schematic on the left in figure five one shows that layer 5 of Cortex projects to spinal pyramidal neurons and this can be interpreted as a prediction okay so this is this one okay helpful um context and and it um relates to this earlier highlighted section which is like it's the interpretation the interpretive link between some anatomical or biological feature phenotype and some parametric resonance with a model the validity of that ranges from pretty clear pretty uncontroversial pretty useful pretty effective to none of the above and it's hard to know like what given interpretive links are doing like saying that this Edge on this base graph can be interpreted as a descending excitatory connection or that might be used to generate specific testable hypotheses like if we measure it during a period of excitation the activity of this neuron we expect it to be increased whereas if it was a descending inhibitory connection Etc that's a great example of active inference being used in a proactive way to generate hypotheses about biological systems that are going to be as they said in the preface answerable to measured data by grounding in the computational model that is related to the neural Anatomy but isn't trying to be like a digital twin of the neuroanatomy we can generate predictions about gain loss of function different measurements to make if they don't already exist and then that can be used to increase our confidence or reliability or falsify even a specific generative model as proposed so there's so much discourse like continuing to the

present day and surely Beyond as like all these questions and recently posted in the Discord about active inference or free energy principle isn't falsifiable now the funny link is it's not falsifiable so it's been falsified it can't be falsified so it's incorrect that's an interesting connection that some people make but a linear model cannot be falsified a base graph cannot be falsified once any given linear model is presented in a context with constraints for a certain data type then it's an empirical question of its accuracy and adequacy relative to other models but at the abstract level of a neural network or linear model or active falsification simply doesn't apply but there's a lot that could be clarified about like what are the utilities and some of the pitfalls of mapping biological or cyber physical systems to active inference models if it's a purely digital system or potentially even a cyber physical system that's been designed a really certain way it might be compatible with actin like essentially by Fiat or by Design one could make an artificial creature like an infra ant that is implementing active inference one could also model an ant in the field using a linear model or a levy flight or some other model or an active inference model but those are two very different settings one in which sort of the rabbit was placed into the Hat and then we have a rabbit analyzer and the other one being like we don't know what's in the Hat but we have a rapid analyzer that we think applies to what is in this hat chapter five includes some um evidence presented on different neurotransmitters so um it'd be interesting to see like what other Frameworks are able to be compatible or even provide unique explanations predictions Etc related to neurochemistry have neural networks ever been used to derive unique predictions or found compatibility with neural hormones neurochemicals and then again they have a final figure in chapter five that's like a graphical overview of several of the systems that are described in the chapter specifically relating to like the cortical cognitive functions the dialectic between habit and free energy driven planning here with dopamine as a Precision modulator with like extensive further modeling presented in other papers and this like even more basal motor selection mechanism based upon reflex arcs and proprioceptive error and all of that we raise these questions not to offer any answers but to highlight some of the exciting Avenues of future research and theoretical neurobiology so a lot of the questions that are asked here in this paragraph but also above at the very beginning they said we're not saying how it is we're just giving like the current process model understanding and then especially at the end they raise even more speculative questions um that takes us to the end of chapter five that's the first half of the book that's the uh epistemic component of the book well a lot of it is epistemic but that's the first five chapters the second five chapters which are longer partially because of having more figures and things like that but you can see

there's more pages um slightly in the second part of the book though we've also read the um at least some of the appendices that's where we're heading which is picking up on uh chapter six in just a few weeks with a recipe for Designing active Emperors models so for those who stuck with the uncertainty for the first five chapters and stay in the game more will become clear when the recipe for the dish is seen and then more will be clarified when the dish is prepared and then when you cut the vegetables and then when you design the dish in the recipe and all of that and that's like a journey that we're all going to be on the tools are not finalized in the kitchen is not completed Etc et cetera et cetera so it's just a call for us to go to onboarding and indicate our interest in continuing with one or the other uh cohort or sharing with any colleagues who you think might like to jump in to the second cohort of part one and then in the coming two weeks we'll be able to take a step back look over the chapter questions but also think about more general questions that we're having basic questions miso questions Advanced questions research Avenues and also turn to the project ideas where multiple of these are already active and there's spaces for people who want to facilitate or catalyze some other direction fun meeting everyone's welcome to uh stay on for tools or head up to a room above if they just want to talk to other people about anything else so thanks again for joining and see you in just a few minutes for tools